

Name: Krish Agrawal
Roll No: 381041
Batch: A2
Assignment No: 6

Problem Statement:

Create a text-to-image generation pipeline using a pre-trained model like DALL-E.

Objective:

- To understand the principles behind text-to-image synthesis using latent diffusion models.
- To configure Stable Diffusion v1.5 via Hugging Face Diffusers with float16 GPU precision.
- To encode a natural language prompt and generate a photorealistic image through iterative denoising.
- To leverage GPU acceleration for fast inference within a practical cloud environment.
- To persist the generated image as a PNG and evaluate its alignment with the input prompt.

S/W Packages and H/W apparatus used:

- Operating System: Windows/Linux/macOS
- Kernel: Python 3.x
- Tools: Google Colab (GPU Runtime - T4)
- Hardware: NVIDIA T4 GPU (16GB VRAM); minimum 12GB VRAM recommended
- Libraries and packages used: Diffusers (StableDiffusionPipeline), Transformers, Accelerate, PyTorch (torch.float16), Matplotlib

Theory:

Text-to-image generation is a multimodal AI capability that synthesises photorealistic or artistic images from natural language descriptions. Contemporary systems use Latent Diffusion Models (LDMs) which operate in a compressed latent space rather than full pixel space, achieving superior image quality at lower computational cost compared to earlier GAN-based or pixel-space diffusion approaches.

DALL-E and Text-to-Image Systems:

DALL-E by OpenAI and Stable Diffusion by Stability AI are the leading text-to-image systems. DALL-E 2 employs CLIP-guided diffusion; Stable Diffusion uses a Latent Diffusion Model with a CLIP text encoder. Stable Diffusion is open-source and deployable via Hugging Face Diffusers, making it the practical implementation choice when API access to DALL-E is unavailable.

Latent Diffusion Models:

A VAE compresses training images into low-dimensional latents. A U-Net denoises progressively within this compact space over T timesteps, guided by cross-attention injection of text embeddings at each step. The VAE decoder reconstructs the final pixel image from the cleaned latent, combining high quality with computational efficiency.

CLIP Text Encoder:

The CLIP encoder converts the input prompt into 77 text embeddings of 768 dimensions each. Pre-trained on 400 million image-text pairs, CLIP aligns text and image representations in a shared semantic space. These embeddings are fed into the U-Net's cross-attention layers at every denoising step to steer synthesis towards the prompt's semantic content.

Classifier-Free Guidance (CFG):

CFG runs two denoising passes per step — one text-conditioned and one unconditional — and combines them as: $\text{guided} = \text{uncond} + \text{scale} * (\text{cond} - \text{uncond})$. A guidance scale of 7.5 strengthens prompt adherence at the cost of some diversity. Stable Diffusion applies CFG by default.

float16 Inference:

Loading with `torch_dtype=torch.float16` halves GPU memory from roughly 7GB to 3.5GB while preserving image quality. NVIDIA Tensor Cores natively accelerate float16 arithmetic, reducing per-step denoising time compared to float32.

Methodology:

- Install `diffusers`, `transformers`, `accelerate`, and `torch`.
- Import `StableDiffusionPipeline` from `diffusers` and `torch`.
- Load `runwayml/stable-diffusion-v1-5` with `torch_dtype=torch.float16` to initialise VAE, U-Net, CLIP encoder, tokenizer, and scheduler.
- Move the pipeline to CUDA via `pipe.to('cuda')`.
- Define the prompt: 'a futuristic robot painting a picture, digital art'.
- Call `pipe(prompt).images[0]` to encode the prompt, initialise a random latent, run 50 guided denoising steps, and decode the result into a 512x512 PIL image.
- Display the image in the notebook.
- Save the output as 'generated_image.png' via `image.save()`.

Use Cases:

- Use Case 1 - Marketing Content Prototyping: Teams rapidly generate campaign visuals, product mockups, and concept artwork from text briefs, reducing reliance on designers for early-stage ideation assets.
- Use Case 2 - Game and Concept Art Production: Studios produce concept art, character sketches, and environment references from text descriptions, accelerating pre-production visual development.
- Use Case 3 - Educational and Scientific Illustration: Educators create custom diagrams, abstract concept visualisations, and historical reconstructions from descriptive prompts.

Advantages:

- Open-source and deployable on consumer GPUs, unlike DALL-E which requires API access.
- Diffusers provides a clean four-line API for the complete generation pipeline.
- float16 halves VRAM usage, enabling inference on a standard cloud T4 GPU.
- Generates diverse, high-fidelity 512x512 images in approximately 7-10 seconds per image.
- PNG outputs are immediately usable in downstream creative workflows.

Limitations:

- Default output is 512x512; higher resolutions require SDXL or post-processing upscalers.

- Complex spatial relationships and precise text-in-image rendering are common failure modes.
- A minimum of 6GB GPU VRAM is required; CPU inference is impractically slow.
- The default pipeline includes an NSFW safety checker that may flag certain prompts.
- Model weights (~5.5GB) require internet access and storage on first download.

Applications:

- Creative content platforms for producing illustrations and concept art.
- E-commerce product visualisation in varied settings and styles.
- Interior design and architecture concept rendering.
- Fashion design: clothing concept renders and textile pattern generation.
- Medical and scientific visualisation for research publications.

Conclusion:

This assignment constructed a complete text-to-image pipeline using Stable Diffusion v1.5 via Hugging Face Diffusers. The model was loaded in float16 on a T4 GPU; the prompt 'a futuristic robot painting a picture, digital art' was encoded by CLIP and processed through 50 guided denoising steps to produce a 512x512 PNG image. The result was displayed and saved for downstream use. The experiment demonstrates how latent diffusion models bridge natural language and visual synthesis, and how Hugging Face has made state-of-the-art text-to-image generation accessible without requiring direct access to proprietary APIs like DALL-E.